

Database Technologies – Assignment 10

PL-SQL

1. Write a procedure that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle, Perimeter = $2*(L+W)$ and Area = $L*W$).

```
KD2_98846_Raj>DELIMITER $$
KD2_98846_Raj>
KD2_98846_Raj>CREATE PROCEDURE sp_rectangle()
-> BEGIN
->
->     DECLARE L INT DEFAULT 20;
->     DECLARE W INT DEFAULT 10;
->     DECLARE perimeter INT;
->     DECLARE area INT;
->
->     SET perimeter = 2 * (L + W);
->     SET area = L * W;
->
->     SELECT L AS length,
->            W AS width,
->            perimeter,
->            area;
->
-> END;
-> $$
```

Query OK, 0 rows affected (0.15 sec)

```
KD2_98846_Raj>
KD2_98846_Raj>DELIMITER ;
KD2_98846_Raj>CALL sp_rectangle();
+-----+-----+-----+-----+
| length | width | perimeter | area |
+-----+-----+-----+-----+
|      20 |     10 |          60 |    200 |
+-----+-----+-----+-----+
1 row in set (0.06 sec)
```

Query OK, 0 rows affected (0.06 sec)

2. Write a procedure that declares an integer variable called num, assigns a value to it, computes and inserts into the temp table the value of the variable itself, its square, and its cube.

```

-> ^C
KD2_98846_Raj>CREATE TABLE temp (
->     num INT,
->     square INT,
->     `cube` INT
-> );
Query OK, 0 rows affected (0.19 sec)

KD2_98846_Raj>DESC temp;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| num   | int  | YES  |     | NULL    |       |
| square | int  | YES  |     | NULL    |       |
| cube  | int  | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.09 sec)

KD2_98846_Raj>DELIMITER $$
KD2_98846_Raj>
KD2_98846_Raj>CREATE PROCEDURE sp_num()
-> BEGIN
->
->     DECLARE num INT DEFAULT 5;
->
->     INSERT INTO temp
->     VALUES(num, num * num, num * num * num);
->
-> END;
->
-> $$
Query OK, 0 rows affected (0.01 sec)

```

```

KD2_98846_Raj>
KD2_98846_Raj>SELECT * FROM temp;
+-----+-----+-----+
| num | square | cube |
+-----+-----+-----+
| 5   | 25    | 125  |
+-----+-----+-----+
1 row in set (0.00 sec)

KD2_98846_Raj>

```

3. Create a procedure to Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are $C = (F - 32) * 5/9$ $F = 9/5 * C + 32$

```
KD2_98846_Raj>DELIMITER $$
KD2_98846_Raj>
KD2_98846_Raj>CREATE PROCEDURE sp_temperature(
  ->     p_temp DOUBLE,
  ->     p_choice INT
  -> )
  -> BEGIN
  ->
  ->     DECLARE v_result DOUBLE;
  ->
  ->     IF p_choice = 1 THEN
  ->
  ->         SET v_result = (p_temp - 32) * 5 / 9;
  ->
  ->         SELECT p_temp AS Fahrenheit,
  ->                v_result AS Celsius;
  ->
  ->     ELSEIF p_choice = 2 THEN
  ->
  ->         SET v_result = 9 * p_temp / 5 + 32;
  ->
  ->         SELECT p_temp AS Celsius,
  ->                v_result AS Fahrenheit;
  ->
  ->     ELSE
  ->
  ->         SELECT 'Invalid choice' AS result;
  ->
  ->     END IF;
  ->
  -> END;
  ->
  -> $$
Query OK, 0 rows affected (0.02 sec)
```

```

KD2_98846_Raj>
KD2_98846_Raj>DELIMITER ;
KD2_98846_Raj>CALL sp_temperature(100, 1);
+-----+-----+
| Fahrenheit | Celsius          |
+-----+-----+
|          100 | 37.77777777777778 |
+-----+-----+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.02 sec)

KD2_98846_Raj>CALL sp_temperature(100, 1);
+-----+-----+
| Fahrenheit | Celsius          |
+-----+-----+
|          100 | 37.77777777777778 |
+-----+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

KD2_98846_Raj>CALL sp_temperature(100, 2);|

```

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

```

KD2_98846_Raj>
KD2_98846_Raj>DELIMITER $$
KD2_98846_Raj>
KD2_98846_Raj>CREATE PROCEDURE sp_inches(p_inches INT)
-> BEGIN
->
->     DECLARE v_yards INT;
->     DECLARE v_feet INT;
->     DECLARE v_inches INT;
->     DECLARE v_remaining INT;
->
->     SET v_yards = p_inches DIV 36;
->
->     SET v_remaining = p_inches MOD 36;
->
->     SET v_feet = v_remaining DIV 12;
->
->     SET v_inches = v_remaining MOD 12;
->
->     SELECT v_yards AS yards,
->            v_feet AS feet,
->            v_inches AS inches;
->
-> END;
->
-> $$

```

Query OK, 0 rows affected (0.02 sec)

```

KD2_98846_Raj>
KD2_98846_Raj>DELIMITER ;
KD2_98846_Raj>CALL sp_inches(124);

```

```

+-----+-----+-----+
| yards | feet | inches |
+-----+-----+-----+
|      3 |     1 |       4 |
+-----+-----+-----+
1 row in set (0.00 sec)

```

Query OK, 0 rows affected (0.00 sec)

5. Create a procedure to read two real numbers and tell whether the product of the two numbers is equal to or greater than 100.

```
KD2_98846_Raj>DROP PROCEDURE IF EXISTS sp_product;
Query OK, 0 rows affected, 1 warning (0.05 sec)

KD2_98846_Raj>
KD2_98846_Raj>DELIMITER $$
KD2_98846_Raj>
KD2_98846_Raj>CREATE PROCEDURE sp_product(
    ->     p_num1 DOUBLE,
    ->     p_num2 DOUBLE
    -> )
    -> BEGIN
    ->
    ->     DECLARE v_product DOUBLE;
    ->
    ->     SET v_product = p_num1 * p_num2;
    ->
    ->     IF v_product >= 100 THEN
    ->
    ->         SELECT p_num1 AS number1,
    ->                p_num2 AS number2,
    ->                v_product AS product,
    ->                'Product is equal to or greater than 100' AS result;
    ->
    ->     ELSE
    ->
    ->         SELECT p_num1 AS number1,
    ->                p_num2 AS number2,
    ->                v_product AS product,
    ->                'Product is less than 100' AS result;
    ->
    ->     END IF;
    ->
    -> END;
    -> $$
Query OK, 0 rows affected (0.02 sec)
```

KD2_98846_Raj>

KD2_98846_Raj>DELIMITER ;

KD2_98846_Raj>CALL sp_product(20.5, 5);

number1	number2	product	result
20.5	5	102.5	Product is equal to or greater than 100

1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

KD2_98846_Raj>CALL sp_product(10, 5);

number1	number2	product	result
10	5	50	Product is less than 100

1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)